# Statistical methods for computer science

Alessandro Cardinali

# Probability

# Where were we?

*Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.*

- ✓ Design phase
- ✓ Description phase

**Today** Introduction to probability

# Course Outline

- Basic concepts
- Descriptive statistics (+ R Lab)
- **Introduction to probability**
- Introduction to random variables
- Discrete probability distributions
- Continuous probability disctributions (+ R Lab)
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

# Introductory concepts

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*Given $\Omega$, an event is a subset of $\Omega$*

Examples: for the fair coin $A = \{H\}$ is an elementary event (composed by a single element)

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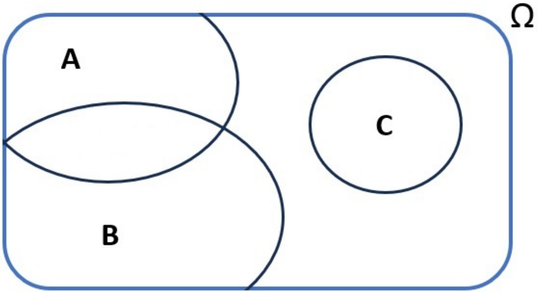
Events can be represented by **Venn's diagrams**. Since events are sets, we can use set's algebra to obtain other events from a starting set of events



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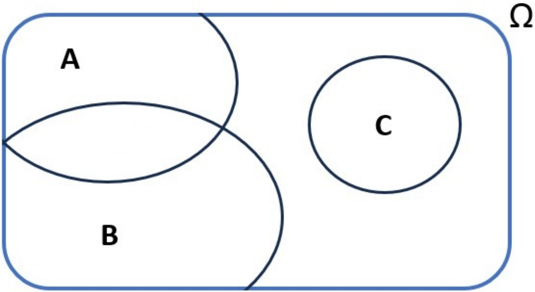


Let's see the main operations...

# Introductory concepts

## Intersection

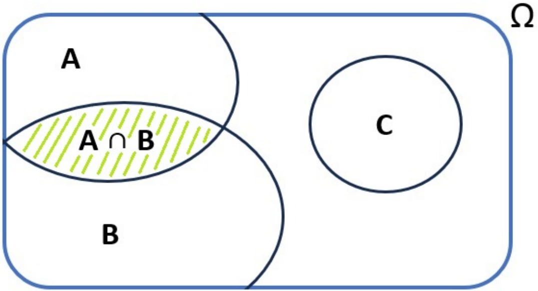


**Figure:** $A \cap B$

Remark: $A \cap C = \varnothing$ (empty set), $A$ and $C$ are called disjoint

# Introductory concepts

## Union

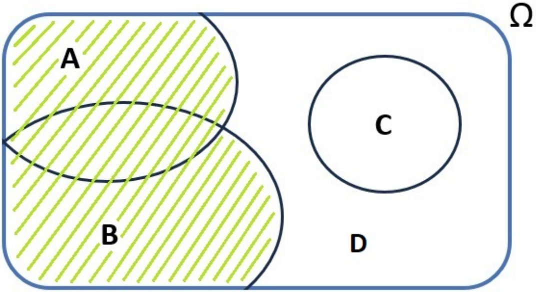


**Figure:** $A \cup B$

Remark: $A \cup B \cup C \cup D = \Omega$ (sure event: at least one between $A$, $B$, $C$ or $D$ will realize)

# Introductory concepts

## Complementary

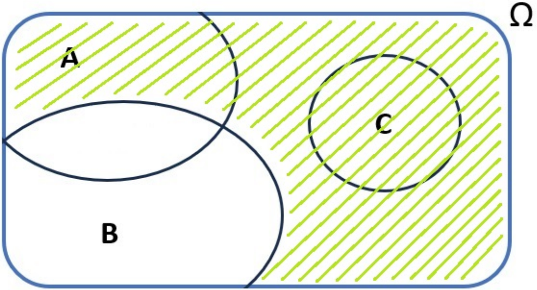


**Figure:** $\overline{B}$

Remarks: $\overline{\Omega} = \varnothing$; $\overline{\varnothing} = \Omega$; $A \cup \overline{A} = \Omega$; $A \cap \overline{A} = \varnothing$

# Introductory concepts

## Set difference

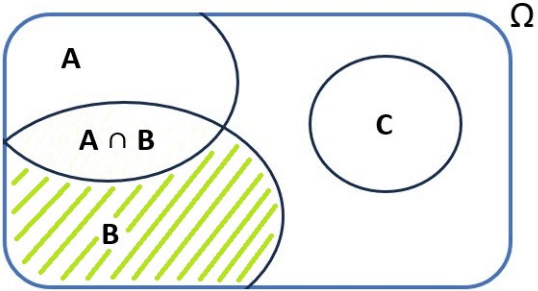


**Figure:** $B \setminus A$

Remarks: $B \setminus A = B \cap \overline{A}$; $B = (A \cap B) \cup (B \setminus A) = (A \cap B) \cup (B \cap \overline{A})$

# Introductory concepts

## Additional remarks

✓ Commutative property

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

✓ Associative property

- $A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C)$
- $A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$

✓ Distributive property

- $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
- $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

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Notes:

- When $\Omega$ is infinite, listing all the elements in $\mathcal{A}$ is impossible, but the important thing to keep in mind is that $\mathcal{A}$ exists
- The event space is also called $\sigma$-algebra and sometimes it is indicated by $\mathcal{F}$

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Consider a fair dice with three faces: $+$, $-$, 0. Build the sample space and the event space

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Let $A = [-50, 100]$, $B = [-60, 20]$, $C = [30, 90]$ and $\Omega = [-100, 100]$. Get $A \cup B$, $A \cap B$, $A \cup C$, $A \cap C$, $B \cup C$, $B \cap C$. Are there any disjoint elementary events?

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- $A \cup B = [-60, 100]$, $A \cap B = [-50, 20]$
- $A \cup C = [-50, 100]$, $A \cap C = [30, 90]$
- $B \cup C = [-60, 90] \setminus (20, 30)$, $B \cap C = \varnothing$ (disjoint)

# Probability

Axiomatic definition of probability (Kolmogorov, 1930)

*Assume we have* $(\Omega, \mathcal{A})$*. The* probability *is a function* $P : \mathcal{A} \longrightarrow \mathbb{R}$ *that to an event* $A \in \mathcal{A}$ *assigns* $P(A) \in \mathbb{R}$ *satisfying*

1. $P(A) \geq 0$, $\forall A$ *(non-negativity)*
2. $P(\Omega) = 1$ *(probability of sure events is 1)*
3. *If* $A$ *and* $B$ *are disjoint,* $P(A \cup B) = P(A) + P(B)$ *(additivity)*

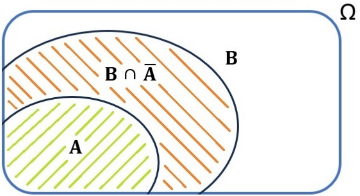
From axioms 1-3 additional properties follow

# Probability

4. $A \subseteq B \Rightarrow P(A) \leq P(B)$

Proof:

$$P(B) = P(A \cup (B \cap \overline{A})) \\ = P(A) + P(B \cap \overline{A}) \geq P(A)$$




5. $P(A) \leq 1$

Proof: it follows directly from 2 and 4, as $A \subseteq \Omega \Rightarrow P(A) \leq P(\Omega)$ and $P(\Omega) = 1$

# Probability

**Remark**

Merging (1) and (5) we have $0 \leq P(A) \leq 1$

6. $P(\overline{A}) = 1 - P(A)$

Proof: From 2 and 3, $P(\Omega) = 1 = P(A \cup \overline{A}) = P(A) + P(\overline{A})$ as $A$ and $\overline{A}$ are disjoint events. It then follows $P(\overline{A}) = 1 - P(A)$

7. $P(\varnothing) = 0$

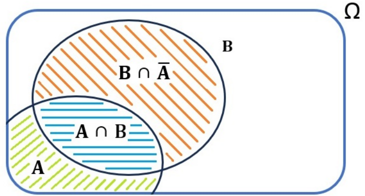
Proof. It follows directly from 6, as $P(\varnothing) = P(\overline{\Omega}) = 1 - P(\Omega) = 0$

# Probability

8. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof:

$$P(A \cup B) = P(A) + P(B \cap \overline{A})$$
$$= P(A) + (P(B) - P(A \cap B))$$




where the last equality follows from the fact that $(A \cap B)$ and $(B \cap \overline{A})$ are disjoint events, thus $P(B) = P(A \cap B) + P(B \cap \overline{A})$

# Probability Calculus

Axiomatic definition of probability:

+ Theoretical foundation of probability theory
– Does not explain how to compute probabilities

To solve real-world problems, it will be necessary to resort to other definitions:

- *Classical* definition
- *Frequentist* definition
- *Subjective* definition

# Classical

*The probability of an event A is equal to the ratio of the number of favorable outcomes ($n_A$) to the total number of possible outcomes ($n$) in a sample space, assuming all outcomes are equally possible, i.e.,*

$$P(A) = \frac{\textit{num. favorable outcomes}}{\textit{num. possible outcomes}} = \frac{n_A}{n}$$

Examples:

- Fair coins, e.g., what is the probability of getting a "head"?
- Fair dice, e.g., what is the probability of getting an even number? what is the probability of the event "getting an even number *or* number 5 "?

# Classical

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Let $A$ be the event "winning Superenalotto", then $n_A = 1$, $n = \binom{90}{6}$ and $P(A) = \frac{n_A}{n} = \frac{6!84!}{90!} = \frac{1}{89 \cdot 87 \cdot 86 \cdot 85 \cdot 11} = 1.6 \cdot 10^{-9}$

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2. $n_A$ for a full house is $13 \cdot \binom{4}{3} \cdot 12\binom{4}{2} = 3744$, so $P(\text{Full House}) = 3744/\binom{52}{5} = 0.001441$

# Classical

Criticism to classical probability:

- Circularity of the definition: asserting that all cases are equally possible means saying that they are equally probable (one cannot define a concept using the same concept)
- Many real-life situations where it is not possible to enumerate favorable and possible cases

  Probability to find 2 bugs in a 200-line code: how many possible outcomes? The possible outcomes range between 0 bugs and ... ?
- The outcomes may not be all equally possible

  Probability of getting an even number when dice are biased (unfair): favorable and possible outcomes can be enumerated but because some sides are heavier than others the probability isn't 3/6

# Frequentist

*The probability of a repeatable event A is the relative frequency of occurrence of the event as the number of trials approaches infinity, assuming that the trials are conducted under identical conditions., i.e., if $n$ is the number of trials and $n_a$ the number of times the event has occurred*

$$P(A) = \lim_{n \to \infty} \frac{n_A}{n}$$

Remarks: some authors interpret the relative frequency of as an approximate measure (for finite $n$) of probability, so they consider probability and relative frequency as the theoretical and empirical sides of the same concept

# Frequentist

**Biased dice**

We know that our die is biased but but we lack information regarding the extent of this bias or its direction. We toss the die 1000 times and record that in 650 instances an even number came up. By definition of frequentist probability, we can now calculate the probability of obtaining an even number as the relative frequency, i.e., $n_A/n = 65\%$

**Bugs**

After collecting 1500 code files of 200 lines each, we observe that 100 of them contain at least 2 bugs. Thus, we can estimate the probability that a 200-line code contains at least 2 bugs as 6.67%.

# Frequentist

Criticism to frequentist probability:

- We can never reach an $\infty$ number of trials, so our probability measures will always be approximations
- Repeatability of trials

  You can imagine to toss a die 1000 times, but what about the probability of being hit by an asteroid?

- Repeatability of trials in unchanged situations: even though we can figure a large number of trials of a repeatable event, the situations are not exactly the same

  The 1500 collected codes should be written by the same person in the same moment

# Subjective

*The probability of an event $A$ is the degree of confidence that a rational individual assigns to the occurrence of an event.*

Remarks:

- It is based on the individual's subjective assessment, knowledge or intuition, rather than empirical data or mathematical calculations
- It solves the theoretical problems of the other two definitions

A computer scientist, based on his/her experience, assigns a probability of 10% to the event "at least 2 bugs in a 200-line code"

Each one of us can also assign a probability to the event "being hit by an asteroid"

# Subjective

Criticism to subjective probability:

- Subjectivity inherent in the definition

  Many different probabilities for the same event

- Difficulty of translating the “degree of confidence” into a meaningful numerical value

# Conditional Probability

You take part in a simple game: one player rolls a fair dice and you win some money if you guess the number that comes up. You'd like to answer number "2".

The probability of the event A "getting 2" is

$$P(A) = 1/6$$

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The probability of $A$ *given* $B$ — or, *conditioning* on $B$ — is $P(A|B) = 1/3$, so it is different (the probability for you to win has doubled!)

# Conditional Probability

*The probability of the event of interest $A$ given that an event $B$ occurred and $P(B) > 0$ is*

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Some useful insights:

- $P(A)$ is equivalent to $P(A|\Omega)$: having no information is equivalent to conditioning on the entire $\Omega$; having some useful information means conditioning on a smaller set (set $B$ in the above example)
- Knowing that $B$ has occurred, the only part of $A$ that may still realize is the intersection $A \cap B$. Thus, $P(A|B)$ is equivalent to reproportionate $P(A \cap B)$ with $P(B)$

Some properties follow from this definition

# Conditional Probability

1. *The conditional probability satisfies axioms 1–3 of the Kolmogorov definition:*
    1. $P(A|B) \geq 0$
    2. $P(\Omega|B) = 1$
    3. $(A_1 \cap A_2) = \varnothing \Rightarrow P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B)$

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1.3 $$P(A_1 \cup A_2|B) = \frac{P((A_1 \cup A_2) \cap B)}{P(B)} = \frac{P((A_1 \cap B) \cup (A_2 \cap B))}{P(B)} = \frac{P(A_1 \cap B) + P(A_2 \cap B)}{P(B)}$$
$$= P(A_1|B) + P(A_2|B)$$

# Conditional Probability

2. *Composite probability formula:*

$$P(A \cap B) = P(A|B)P(B)$$

## Remark

As the intersection between sets is commutative, it holds $P(A \cap B) = P(A|B)P(B) = P(B \cap A)$ and by the conditional probability formula $P(B|A) = \frac{P(B \cap A)}{P(A)}$, thus $P(B \cap A) = P(B|A)P(A)$. It then follows: $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

# Conditional Probability

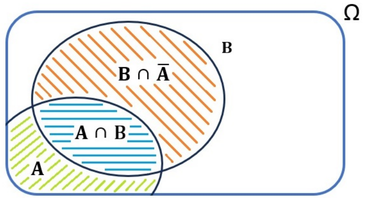
3. *Marginal probability formula:*

$$P(B) = P(B|A)P(A) + P(B|\overline{A})P(\overline{A})$$

Proof:

$$\begin{aligned} P(B) &= P(B \cap A) + P(B \cap \overline{A}) \\ &= P(B|A)P(A) + P(B|\overline{A})P(\overline{A}) \end{aligned}$$

where the last equality follows from the composite probability formula



# Conditional Probability

4. *Bayes formula:*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})}$$

- Named after Thomas Bayes, an XVIII century English reverend and statistician
- This simple rule provided the foundations for **Bayesian inference**, an entirely different approach to statistical inference
- Under the Bayesian inference interpretation, this rule expresses how to formally update our prior belief over A (an unknown parameter) when new information B comes in (the observed data); $P(A)$ and $P(A|B)$ are then called, respectively, *prior* and *posterior* probabilities

# Example

Calculate the new probability of winning the dice game after the disclosure that an even number was rolled (hint: use Bayes formula)

$A$: "rolling a 2 from a fair dice", $P(A) = 1/6$
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$B|A$: "getting an even number, given that 2 was rolled", $P(B|A) = 1$

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$A$: "rolling a 2 from a fair dice", $P(A) = 1/6$
$B$: "rolling an even number from a fair dice", $P(B) = 1/2$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

$B|A$: "getting an even number, given that 2 was rolled", $P(B|A) = 1$

$$P(A|B) = \frac{1 \cdot 1/6}{1/2} = 1/3$$

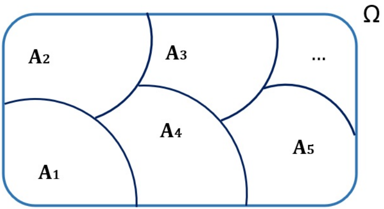
# Conditional Probability (Generalized)

The properties 2–4 can be generalized to multiple events. To do that, we first need to introduce the concept of partition

*The events $A_1, A_2, \ldots, A_n$ are a partition of $\Omega$ if and only if*

- $A_i \cap A_j = \varnothing$ *for all* $i \neq j$ *(all events are disjoint)*
- $A_1 \cup A_2 \cup \cdots \cup A_n = \Omega$

Note: $A$ and $\overline{A}$ are a partition of $\Omega$ (presumed by properties 2 and 3)



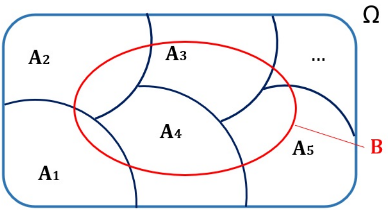
# Conditional Probability (Generalized)

**2G.** *Composite probability formula (generalized):*

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdot \ldots \cdot P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1})$$

**3G.** *Marginal probability formula (generalized):*

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_n)P(A_n)$$




Property 2G is also called "chain rule". The proofs of 2G and 3G are analogous to the special case with two events

# Conditional Probability (Generalized)

**4G.** *Bayes formula (generalized):*

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^{N} P(B|A_i)P(A_i)}$$

- $P(B)$ at the denominator is usually computed using the marginal probability formula
- $P(A_i)$ is named *prior* probability
- $P(B|A_i)$ is named *posterior* probability, as it is the update of $P(A_i)$ given the new information $B$

# Independence

*Events $A, B$ are independent if and only if*

- $P(A|B) = P(A)$, *or*
- $P(B|A) = P(B)$, *or*
- $P(A \cap B) = P(A)P(B)$

- This means that knowing one of the two events gives no useful information (in terms of probability) on the other event
- If one of the conditions highlighted above holds, the other two conditions hold as well
- The "if and only if" connection means that the definition holds in both directions

# Exercise

Define the events $A, B$ such that $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/6$.

- Are $A, B$ independent?

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Define the independent events $A, B$ such that $P(A) = 1/2$, $P(B) = 1/4$

- $P(A \cap B)$ ?

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Define the events $A, B$ such that $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/6$.

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Define the independent events $A, B$ such that $P(A) = 1/2$, $P(B) = 1/4$

- $P(A \cap B)$ ? By definition of independent events, $P(A \cap B) = P(A) \cdot P(B) = 1/8$
- $P(A|B)$ ?

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